

TO SUPPLY LEFTOVER FOOD TO THE POOR

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Team ID	LTVIP2025TMID29707
Project Name	To Supply Leftover food to Poor
Maximum Marks	

Chapter 2 Ideation Phase

2.1 Problem Statement:

Despite significant advancements in food production and distribution, **India wastes approximately 67 million tons of food annually**, valued at over ₹1 lakh crore. This paradox exists alongside widespread hunger and malnutrition, especially among urban homeless populations and low-income communities. Restaurants, event venues, and households often discard surplus food due to a lack of awareness, logistical challenges, and the absence of a streamlined donation system.

There is **no unified platform** that connects food donors with NGOs or volunteers in real time, resulting in missed opportunities to redistribute edible food. Additionally, concerns about food safety, transportation, and coordination further hinder efforts to repurpose leftovers.

This project aims to develop a **digital solution** that bridges the gap between food donors and recipients, enabling efficient, safe, and timely redistribution of surplus food. By leveraging technology, the system will reduce food waste, alleviate hunger, and promote social responsibility.

2.2 Empathy Map Canvas:

☐ Who are we empathizing with?

- **Primary User:** Volunteer or NGO worker distributing food
- **Secondary Users:** Food donors (restaurants, event organizers), recipients (homeless, low-income families)

☐ What do they do?

- Volunteers coordinate pickups and deliveries
- Donors pack leftover food
- Recipients queue at distribution points or shelters

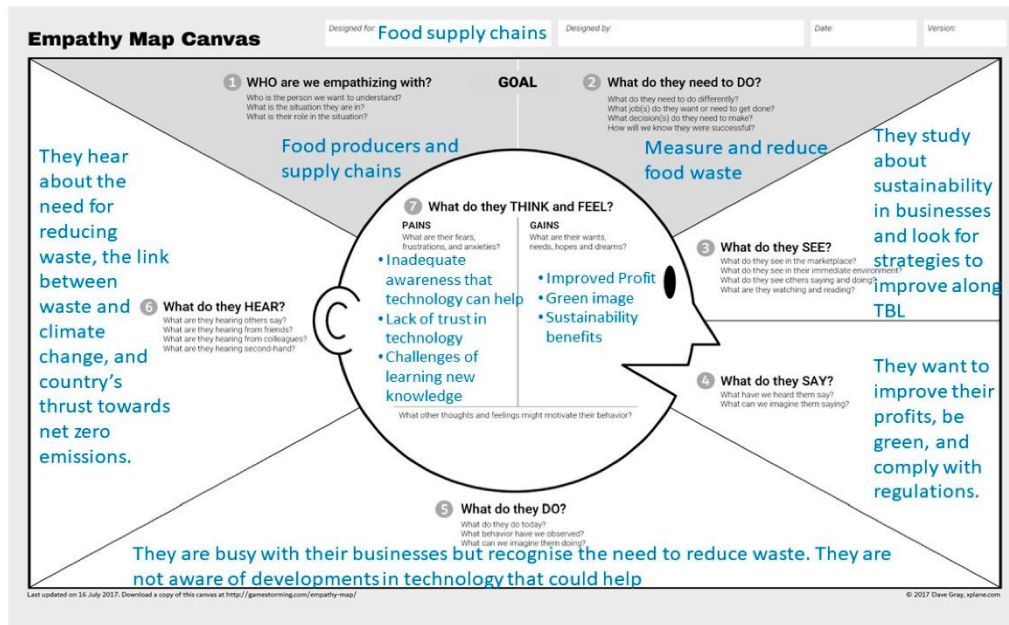
☐ Pains

- Unclear logistics and coordination
- Food spoilage due to delays
- Limited awareness among donors
- Lack of trust in food safety

☐ Gains

- Reduced food waste
- Feeding the hungry
- Community engagement
- Sense of purpose and impact

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2.3 Brainstorming:

☐ Core Features to Build

- **Donor App/Portal:** Restaurants, caterers, and households can list surplus food with pickup time and quantity.
- **Volunteer Dashboard:** Shows nearby donations, routes, and delivery status.
- **Recipient Locator:** Maps shelters, slums, and food-insecure zones for targeted distribution.
- **Food Safety Module:** Includes expiry tracking, hygiene guidelines, and quality checks.
- **Impact Tracker:** Visualizes meals served, food saved, and carbon footprint reduced.

☐ Creative Add-ons

- **Smart Bins:** IoT-enabled bins at event venues that notify volunteers when filled with edible food.
- **AI Routing:** Optimizes pickup and delivery routes based on traffic and urgency.
- **Gamified Volunteer System:** Points, badges, and leaderboards to encourage participation.
- **Community Fridges:** Public-access fridges in neighbourhoods stocked with donated food.
- **Voice Alerts:** For donors or volunteers with limited literacy—simple voice-based instructions.

☐ Tech Stack Ideas

Layer	Tools/Tech Suggestions
Frontend	Flutter, React Native
Backend	Node.js, Firebase, Django
Database	Firestore, MongoDB
Notifications	Twilio (SMS), Firebase Cloud Messaging
Hosting	Heroku, AWS, Vercel

☐ Stakeholders to Engage

- Local NGOs and food banks
- Municipal authorities for permissions

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- Restaurants, hotels, and event venues
- Student volunteers and social groups
- Health departments for food safety compliance

